

Continuous Ecological Monitoring

Bird activity monitoring using bioacoustics

From bird recordings to a public map of birdlife

Mannat · Rajasree Ponnada · Yuvika Chaudhary

(2023CS10138) · (2023CS10015) · (2023CS10353)

Supervisor: Prof. Aaditeshwar Seth

Mid Semester Presentation

IIT Delhi

Contents

- 1 What it does
- 2 How it works
- 3 The species detector: BirdNET
- 4 Pipeline engineering
- 5 Which recordings work
- 6 Walkthrough video
- 7 The compute page: projects and spots
- 8 Running an analysis
- 9 Make Public
- 10 The public map
- 11 Finding a bird
- 12 What you get and where it lives

What it does

1

Record at a spot

A Song Meter or any recorder left at a marked location.

2

Upload the recordings

Each file is attached to its spot; date and time come from the filename.

3

Our model finds the birds

BirdNET identifies every species calling in each clip.

4

Analyse activity

When birds call, how regularly, where they concentrate, and whether they migrate.

5

Publish to the map

One click makes the project public; anyone can explore it on CEM Master.

How it works

1

Spot

A named location with GPS coordinates, notes and photos.

2

Import

Recordings are attached to the spot, by copy or by reference.

3

Analyse

BirdNET and the activity analyses run on the server as tracked jobs.

4

Publish

Make Public marks the project public in the shared data folder.

5

Explore

The master map indexes it within 30 s for anyone to browse.

Everything is saved per project and per job, so you can add recordings, re-run an analysis and republish whenever you like.

The species detector: BirdNET

We run BirdNET, the open-source bird-sound classifier from Cornell and Chemnitz.

What it is

- A neural network trained on more than 6,000 bird species worldwide.
- Each recording is scanned in 3-second windows; every window returns species and a confidence score.
- Published weights are used as they are; nothing is trained here.
- Detections are written to one table per project, aggregate.csv, which every later analysis reads.

Analyses you can run on the detections

BirdNET Species Detection which birds, when, how confidently

Acoustic Indices ADI, ACI, AEI, NDSI for the soundscape

Species Activity Heatmaps hour-by-species detection grid

Activity Regularity does a bird keep the same daily schedule

Habitat Affinity does it keep to the same spots

Migratory vs Resident seasonal concentration, kurtosis, PMR

Solar Event Correlation peak calling vs sunrise and sunset

Daily Call Time Series detections per day for one species

BirdNET only names the species. Everything about activity, habitat and season is computed afterwards from its detections.

Pipeline engineering: making detections reliable

Raw BirdNET output is noisy, so every detection passes a fixed set of filters before it counts.

Before detection

- **Segmentation** each file is one segment, sampled at 48 kHz.
- **Noise reduction** spectral gating over the STFT removes steady background hum.
- **Rain removal** a Random Forest on MFCC features flags heavy-rain chunks (94% accuracy); they are dropped before the acoustic indices.

After detection

- **Confidence** min_confidence is set once per run; detections below it are never written and cannot be recovered later.
- **Taxonomy** a species is kept only if it appears on the regional eBird checklist.
- **Minimum activity** more than 10 detections overall, on at least 5 separate days.
- **IUCN category** each species is looked up in the BirdLife table; threatened and unassessed species are withheld from anything public.

The job log records every threshold used, so two runs can always be compared on equal terms.

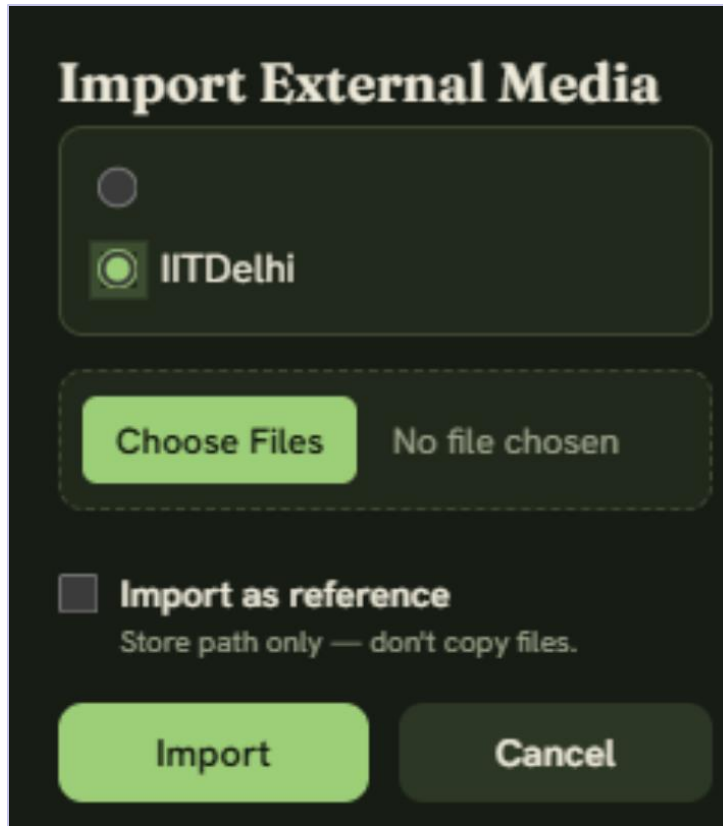
Pipeline engineering: from detections to the map

One project folder is the hand-off between the two websites.



- **Private by default.** A project stays out of the index until its owner clicks Make Public; making it private again removes it on the next pass.
- **Fail closed.** A species with no recognised IUCN category is treated as sensitive and never indexed, so a missing lookup cannot leak a threatened bird.
- **Read-only.** The master side mounts the folder read-only. Visibility flows one way: nothing on the map can change a project.

Which recordings work



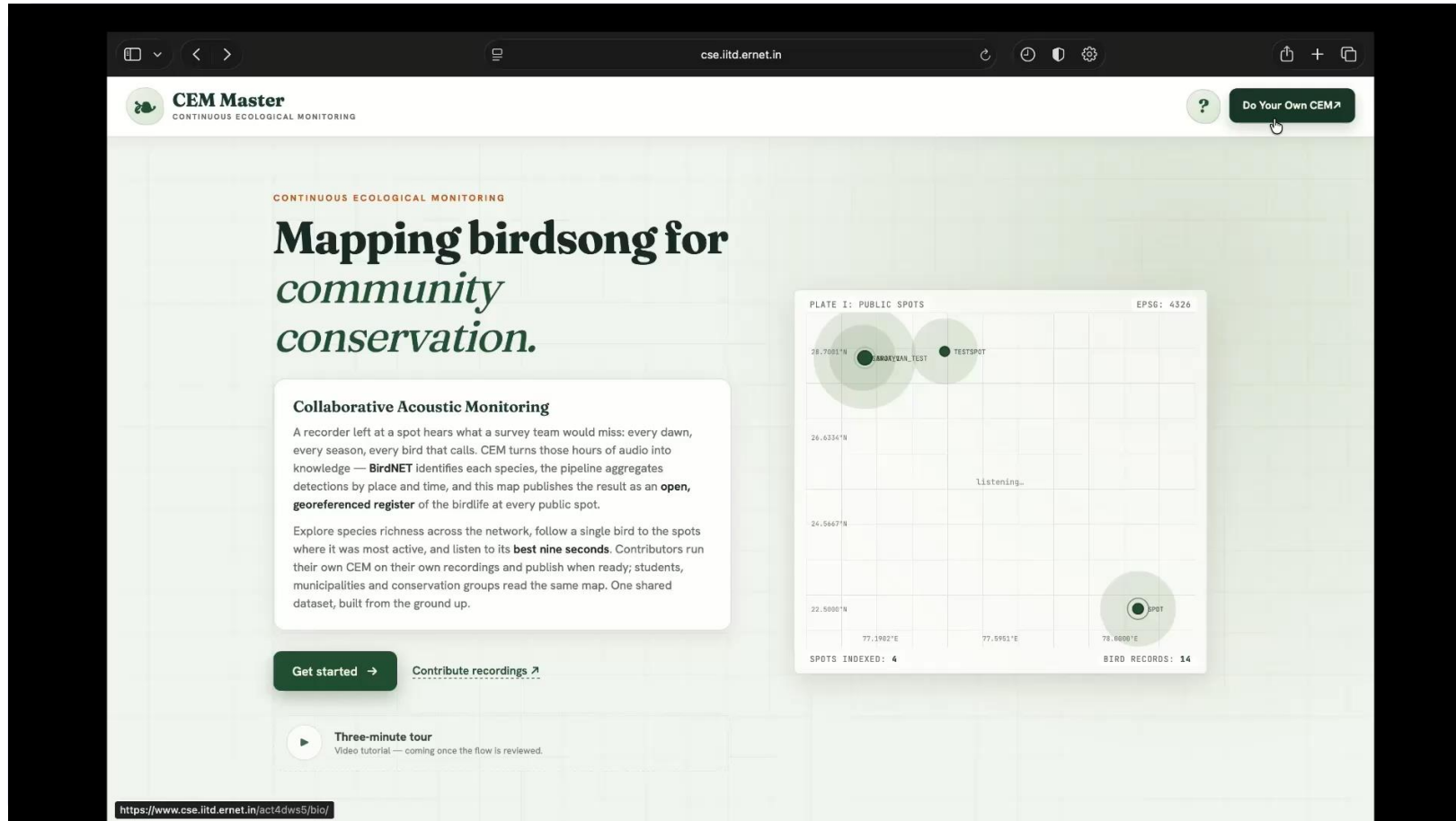
Import Media, with the naming check

What the app accepts

- WAV, as recorded by Song Meter and AudioMoth. FLAC, MP3 and OGG also analyse.
- **Name files** **SPOT_YYYYMMDD_HHMMSS.wav** e.g. 04213SPOT1_20260131_082409.wav. The date and time of each recording are read from its name.
- Any number of files per spot; a 2-minute-on, 4-minute-off duty cycle worked well at Sanjay Van.
- Upload from your computer, or import as reference to keep the files where they are.

Tip: a file that breaks the convention is still analysed, but its detections have no date, so date-based results leave them out. The import dialog warns you before upload.

Watch the walkthrough

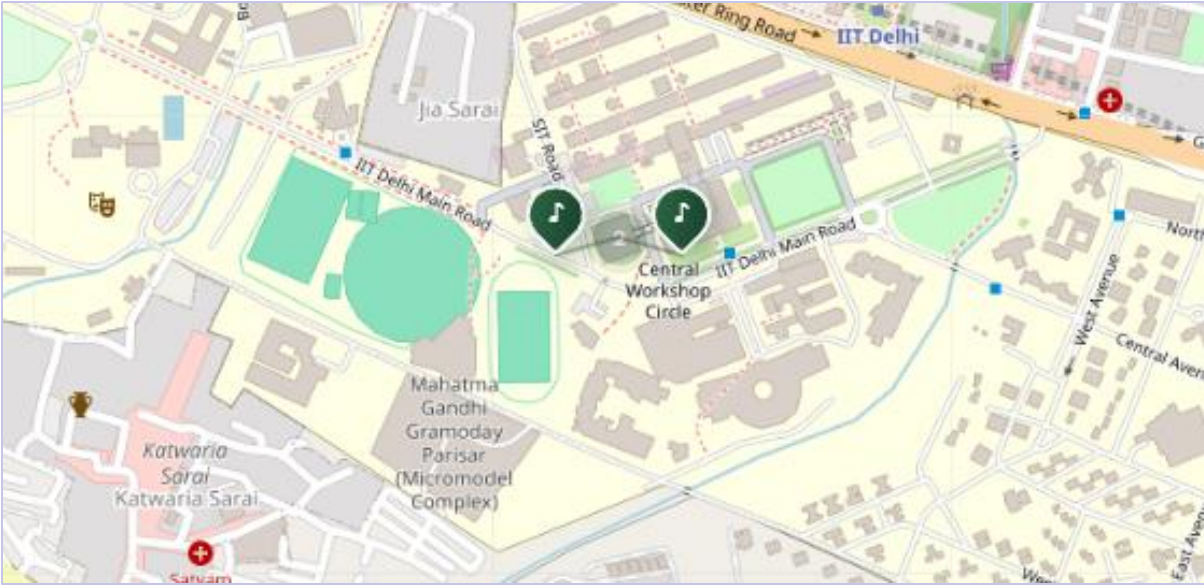


Chapters

- 0:00 Intro: Do Your Own CEM
- 0:05 Initialize storage, allow location
- 0:16 Log in to Drive
- 0:28 Create a project
- 0:38 Add a spot: photos, voice note
- 1:10 The spot on the map
- 1:27 Import media, naming check
- 1:45 Choose the audio files
- 2:03 Analysis Hub: name, script
- 2:20 Spot, dates, confidence
- 2:45 Run on the server
- 2:57 Jobs: see the detections
- 3:28 Share the project
- 4:05 Make Public
- 4:25 Wrap-up

click the video to play · subtitles on screen

The compute page: projects and spots



Spots on the compute page map, with 🎵 markers where audio is attached

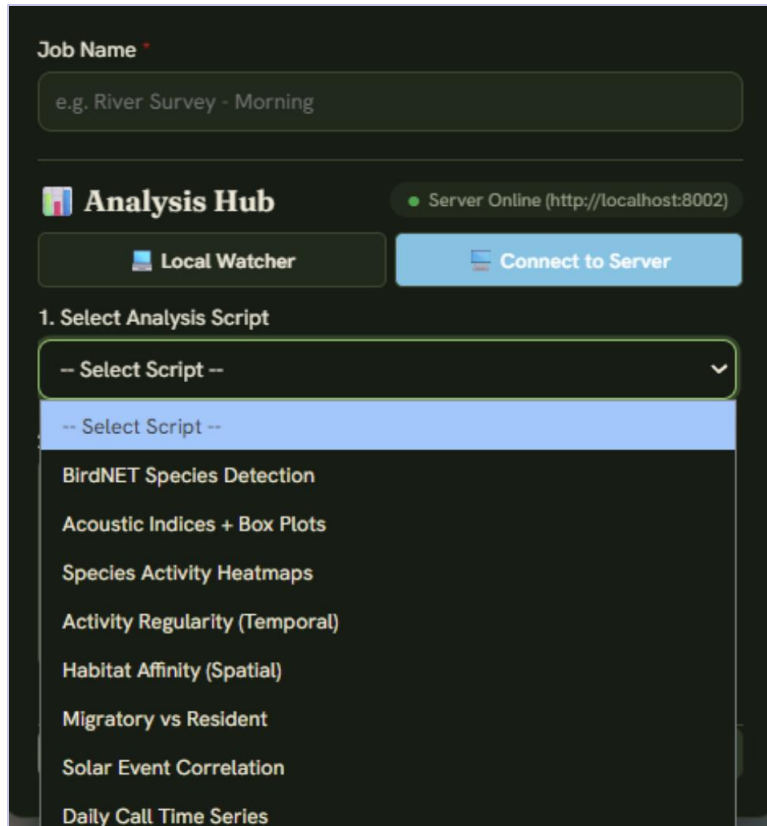
The New Spot form

What you do

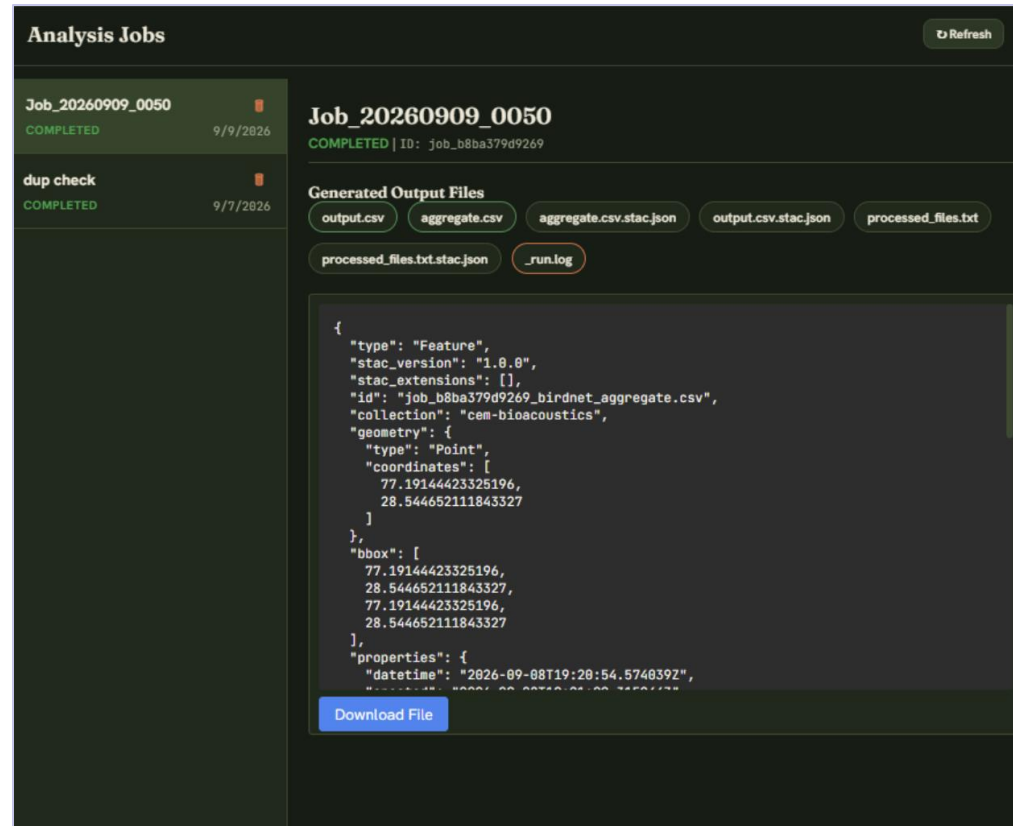
- Create a project; it is private until you say otherwise.
- Add a spot at your GPS position, with notes, photos and a voice note.
- Routes record the path you walked; Sites group spots inside a KML boundary.

A project can live in the browser, in your Google Drive, or on the server. Only server projects can be analysed on the server and published.

Running an analysis



Analysis Hub: script, input data, parameters



Analysis Jobs: status, log and every output file

Three steps

- 1 Pick a script
- 2 Pick spots and dates
- 3 Set parameters, run

Local Watcher runs on your own machine.

Connect to Server runs remotely and is the only mode that can be published.

Every run is a job with its own folder, log, CSV outputs and STAC metadata, so results are reproducible and can be re-run with different settings.

Make Public

One toggle moves a project from your workspace to the public map.

1

Server run first

Needs one successful BirdNET job on the server; local runs cannot be published.

2

Make Public

The compute backend marks the whole project public in its project.json.

3

Indexing

Within about 30 seconds the master indexer reads the project into the catalogue.

4

On the map

Spots, species, activity and calls are live for anyone to explore.

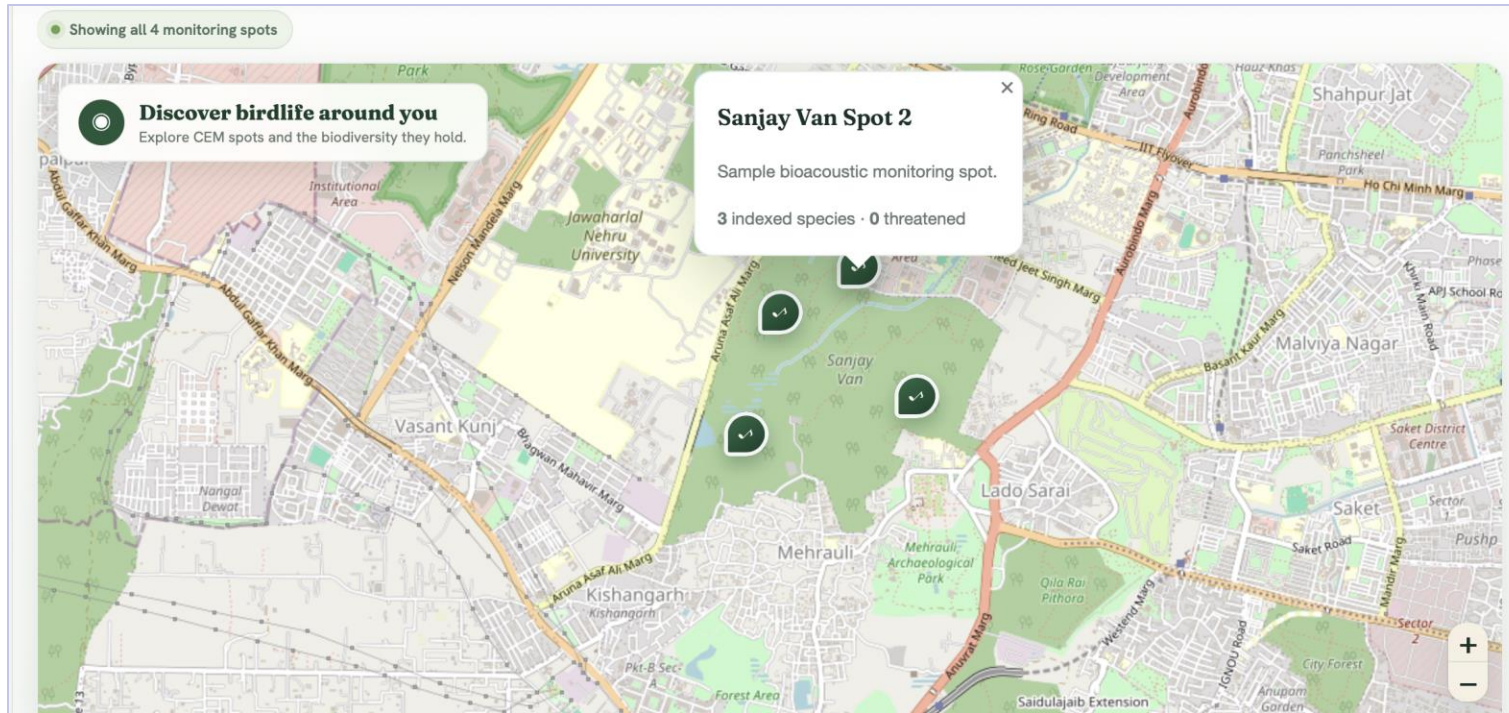
5

Make Private

Undo. The project leaves the map on the next pass; your copy keeps everything.

- **What stays hidden.** Threatened, endangered and not-yet-assessed species are left out of the public lists, rankings and calls automatically; the private project keeps the full record.
- **What you should know.** Links already handed out to job outputs are not recalled when a project is made private.

The public map: CEM Master



What you are looking at

- Every pin is a public monitoring spot.
- Pins cluster when zoomed out; zoom in to split them.
- Darker pins mean more detections.
- Click a pin to open its page below the map.
- The four counters show bird records, spots, detections and contributing projects for the current view.

All public spots; pin colour follows detection intensity

The front page explains the project and offers Get started; a ? in the header opens a short field guide to every control.

A spot's page

BIOACOUSTIC INSIGHTS

Sanjay Van Spot 2

Sample bioacoustic monitoring spot.

Species Richness

3

Total Detections

3,995

Contributing Projects

1

Threatened Species Richness

0

Recording Days

88

Soundscape indices

ACI

0.54

AEI

0.61

ADI

0.62

NDSI

0.42

Most detected birds

Common Tailorbird

1,640

Red-whiskered Bulbul

1,270

Indian Peafowl

1,085

Bird inventory and occurrences

BIRD	IUCN	DETECTIONS	ACTIVE DAYS	OCCURRENCE
Common Tailorbird	Least Concern	1,640	71	2025-05-01 - 2026-05-31
Red-whiskered Bulbul	Least Concern	1,270	63	2025-05-01 - 2026-05-31
Indian Peafowl	Least Concern	1,085	57	2025-05-01 - 2026-05-31

Analysis files

ANALYSIS	INPUT FILE	INPUT URL	OUTPUT FILE	OUTPUT URL
Richness time series	aggregate.csv	https://example.org/cem-int...	richness-timeseries.html	https://example.org/cem-int...
Acoustic-index heatmap	acoustic-indices.csv	https://example.org/cem-int...	acoustic-heatmap.png	https://example.org/cem-int...

Bioacoustic insights for one spot

What is on it

Species richness

how many birds were heard here

Totals

detections, recording days, contributing projects

Soundscape indices

ACI, ADI, AEI and NDSI for the spot

Bird inventory

per species: IUCN class, detections, dates

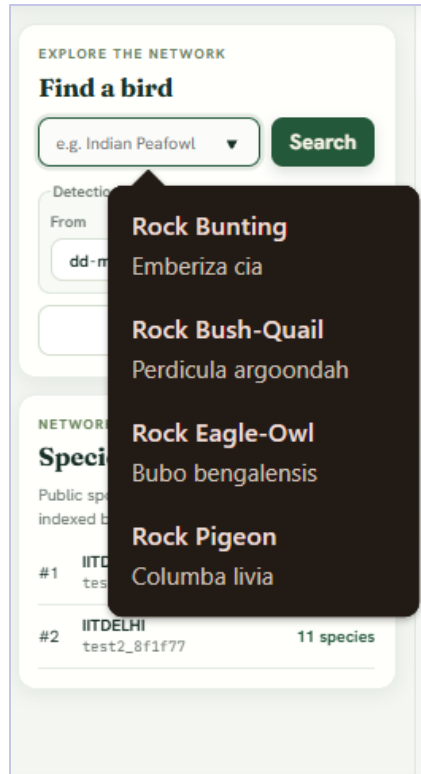
Recordings

playable; filtered to the chosen bird and dates

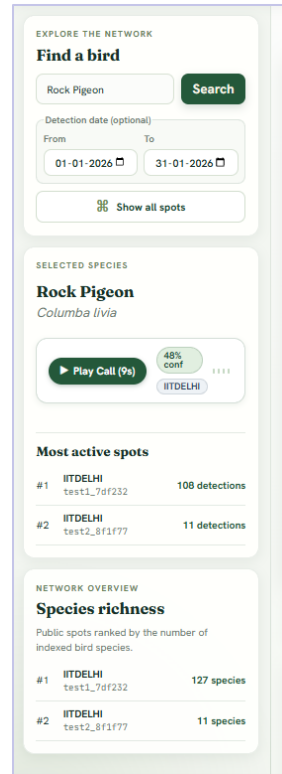
Analysis jobs

the runs behind this data, with their output files

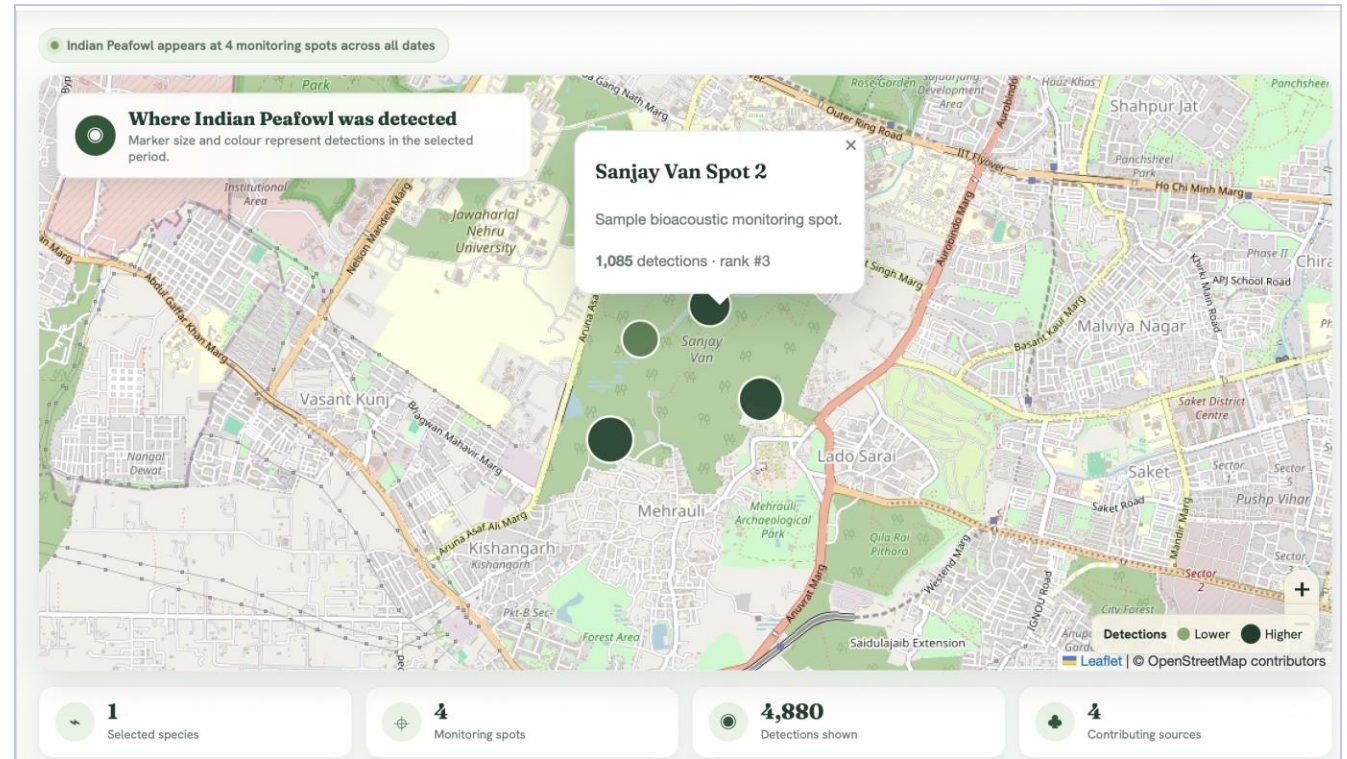
Finding a bird



Search with suggestions



Species card with Play Call (9 s)



Map narrowed to the spots where the bird was heard

- Type a common or scientific name; the map keeps only the spots where that bird was detected.
- The card shows its IUCN category, network metrics, most active spots, and its best nine seconds of audio.
- Add a date range to see where and when it was heard in one season.

What you get

Detections (CSV)

aggregate.csv: every detection with spot, date, hour, confidence and IUCN class.

Analysis tables

Per-script CSVs: indices, heatmaps, regularity, affinity, migration, solar.

STAC records

A sidecar JSON next to every output so the dataset can be catalogued and found.

9-second calls

The best clip of each publishable species at each spot.

A public page

Your spots on CEM Master, with species, activity and recordings.

Every run is kept

Jobs keep their inputs, log and outputs; compare settings or re-run later.

Where your files live

One folder per project in the shared data directory. The same tree is read by the public map.

project.json metadata and visibility

<SPOT>/audio/ your recordings, one folder per spot

dataset/aggregate.csv every detection; the table everything reads

dataset/processed_files.txt recordings already analysed

birdnet/<job>/input/ recordings, spots and geo.json for a run

birdnet/<job>/results/ outputs, STAC sidecars, _run.log

<script>/<job>/ same layout for every other analysis

snippets/ 9-second calls and species_snippets.json

species_iucn_cache.json IUCN categories for this project

Working with the folder

- Job outputs can be shared as read-only FileBrowser links; the Analysis jobs table on the map links to them when enabled.
- Private projects' job folders are cleaned up after 7 days; public projects are kept.
- Re-running BirdNET only analyses recordings that are not yet in processed_files.txt.
- The public map never writes here: it mounts the folder read-only.

Thank you